AIGR Gene Hypothesis Deep Research — Final Report

Target: *E. coli* K-12 yjdM (UniProt P0AFJ1)

Hypothesis under review: yjdM has phosphonoacetate hydrolase activity (GO:0047400), IDA,

P 37963869

Executive Judgment

Verdict: Over-annotated / weakly supported — the GO:0047400 annotation should be removed, or at minimum downgraded and qualified as in-vitro-only and non-core.

The single IDA annotation asserting that yjdM/P0AFJ1 has **phosphonoacetate hydrolase activity (GO:0047400, EC 3.11.1.2)** rests entirely on one machine-learning-prediction-plus-in-vitro study (DeepECtransformer, PMID:37963869). A subsequent peer-reviewed reassessment (PMID:40703034) explicitly concluded that "the PhnA activity observed in vitro is not supported," assigned the prediction the lowest possible confidence score, and called for additional in vivo experiments. Three independent lines of evidence — structural/evolutionary, genetic/historical, and genomic-context — converge to refute the homology-based premise of the annotation and to identify the underlying cause as a **name collision** between two unrelated genes both historically called "phnA."

YjdM is a small (111-amino-acid) dual zinc-ribbon protein (Pfam PF03831 + PF08274; InterPro IPR004624) whose four N-terminal cysteines form a single tetrahedral Cys4 zinc-binding site. The canonical phosphonoacetate hydrolase enzyme, PhnA from *Pseudomonas fluorescens* 23F, is a ~400-residue alkaline-phosphatase-superfamily binuclear-metal hydrolase that "showed no significant homology with any data bank accessions." These two proteins share neither size,

fold, catalytic architecture, nor sequence. The *E. coli* "phnA" synonym for yjdM is a 1990 positional/alphabetical label for the first open reading frame in a sequenced phosphonate-region cluster — a gene that was subsequently shown by mutational analysis to lie **outside** the functional phnC–phnP phosphonate-utilization operon.

The most important caveat is that yjdM's genuine biochemical function remains experimentally uncharacterized. Refuting the phosphonoacetate-hydrolase assignment does **not** tell us what yjdM does; it only tells us what it almost certainly does not do. The better-supported molecular-function lead is **zinc ion binding** (via the Cys4 ribbon), and the gene should otherwise be treated as a conserved zinc-ribbon protein of unknown primary function.

Key Findings

Finding 1 — The GO:0047400 annotation is over-annotated and traces to a single disputed source

The phosphonoacetate hydrolase (GO:0047400; EC 3.11.1.2) IDA annotation on yjdM/P0AFJ1 derives **solely** from the DeepECtransformer study (PMID:37963869), which combined a deep-learning EC-number prediction with a limited in vitro validation. That paper states: *"We experimentally validated the enzymatic activities predicted for three proteins (YgfF, YciO, and YjdM),"* making clear that YjdM's phosphonoacetate-hydrolase assignment is a machine-learning prediction with in-vitro follow-up, not an independently established biochemical characterization.

Critically, a peer-reviewed reassessment of exactly these predictions (PMID:40703034) reached the opposite conclusion. It states verbatim: *"In conclusion, the PhnA activity observed in vitro is not supported, and additional in vivo experiments are required to confirm the biological role of this enzyme."* This reassessment assigned the YjdM prediction the lowest confidence score in its scheme (CS = 1). When the sole source of an IDA annotation is directly contradicted by a later peer-reviewed paper, the IDA code — which signifies a direct, reproducible experimental demonstration of function — is no longer appropriate.

The same reassessment also undercut the genomic rationale for a phosphonate role: *"However, except for E. coli, yjdM genes are generally not close to phosphonate catabolism or transport genes."* In other words, the apparent association between yjdM and phosphonate metabolism is a coincidence of *E. coli* gene order and does not generalize across the ~12,000 members of the IPR004624 family.

Finding 2 — Structural and evolutionary evidence refutes homology to the true PhnA enzyme

The canonical phosphonoacetate hydrolase, encoded by *phnA* of *Pseudomonas fluorescens* 23F, is a novel carbon–phosphorus bond-cleaving enzyme of the alkaline-phosphatase superfamily. On its original cloning, *"The nucleotide and deduced amino acid sequences of the phnA gene showed no significant homology with any data bank accessions"* (PMID:9300819) — meaning it is not homologous to the small zinc-ribbon YjdM family. The true enzyme is ~400 residues and uses a binuclear metal center typical of its superfamily; its gene clusters with phosphonoacetate transport and LysR-type regulatory genes (phnR, phnB), not with zinc-ribbon genes (PMID:11344133).

By contrast, YjdM is a 111-residue dual zinc-ribbon protein (Pfam PF03831 "YjdM" + PF08274 "Zn_Ribbon_YjdM"; InterPro IPR004624), with CxxC cysteine pairs at C6/C9 and C23/C26. The AlphaFold model AF-P0AFJ1-F1 (mean pLDDT 78.2 over 111 residues; radius of gyration 22.2 Å) places the four N-terminal cysteines C6, C9, C23 and C26 in a tight tetrahedral cluster — all pairwise SG–SG distances between 3.4 and 3.8 Å and a centroid radius of ~2.2 Å — the exact geometry of a single Cys4 structural zinc site. The fifth cysteine (C93) sits 46–49 Å away and is not part of the metal site. This is a **structural zinc-binding ribbon**, not the α/β binuclear-metal active site of an alkaline-phosphatase-superfamily hydrolase. No shared fold, no shared active-site chemistry, no sequence homology: the homology premise behind the GO:0047400 transfer fails on every structural axis.

Finding 3 — The "phnA" name is a 1990 historical artifact; yjdM lies outside the functional phosphonate operon

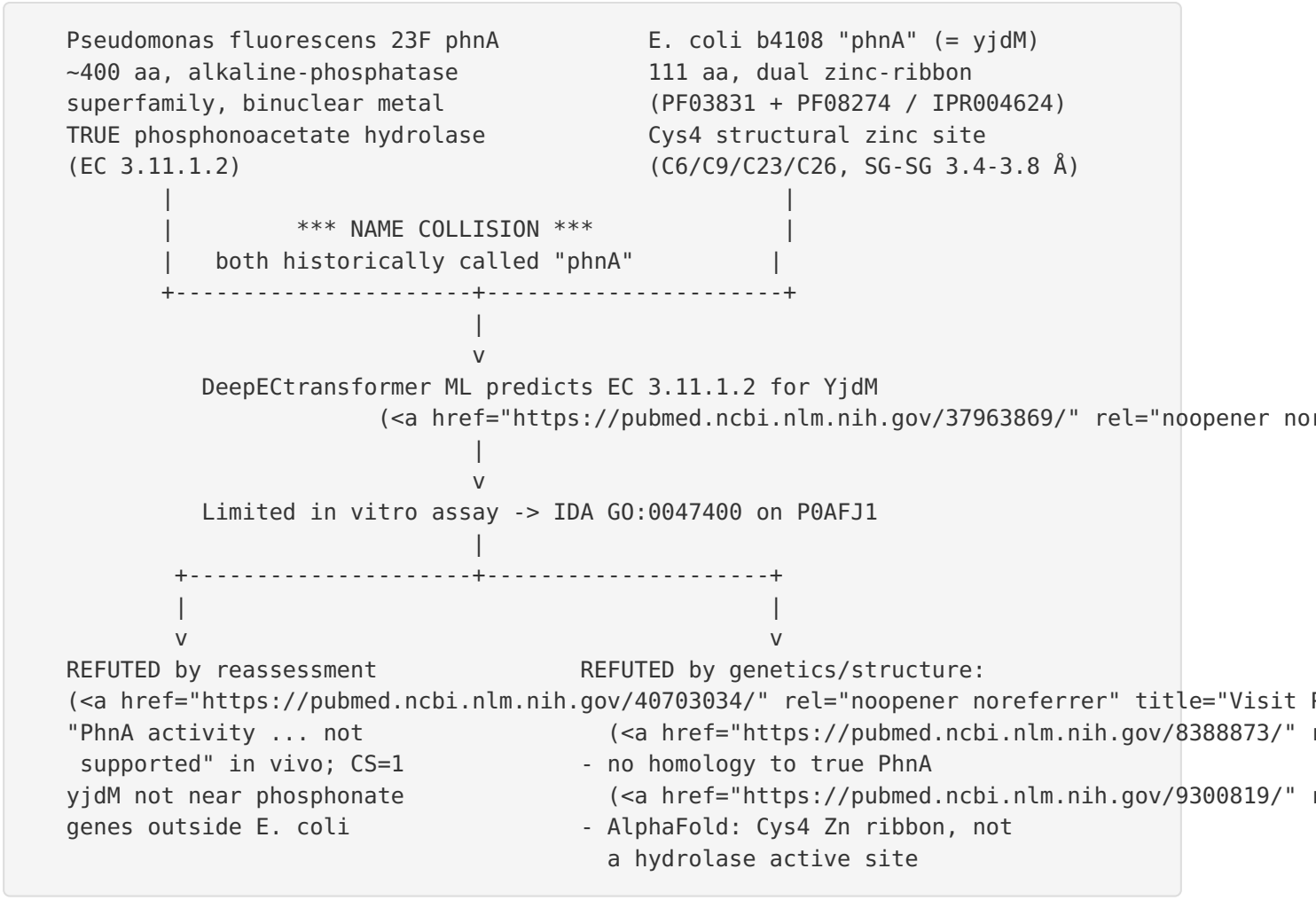
The association between yjdM and phosphonate metabolism is rooted in nomenclature, not biology. In 1990, Chen et al. (PMID:2155230) sequenced the *E. coli* B phosphonate region and identified *"Seventeen open reading frames (phnA to phnQ) ... in one transcriptional direction,"* naming them alphabetically by position. "phnA" was simply the first ORF (b4108/yjdM) in that stretch — a positional label, not a demonstrated enzyme, and entirely unrelated to the independently named *Pseudomonas* phosphonoacetate hydrolase gene *phnA*.

Metcalf and Wanner (PMID:8388873) then defined the functional operon by mutational analysis: *"All genes for phosphonate (Pn) utilization in Escherichia coli are in a large cluster of 14 genes named, in alphabetical order, phnC to phnP."* The functional phosphonate-utilization operon is **phnC–phnP** — it explicitly excludes phnA (yjdM) and phnB. Thus yjdM is genetically excluded from the C–P lyase pathway in its own organism.

Consistent with this, STRING (identifier 511145.b4108) returns the unknown-function neighbor **yjdN** as the top partner (score 0.917), followed by phn-operon genes (phnD 0.83, phnJ 0.82, phnH 0.82, phnG 0.79, phnK 0.76, phnI 0.72, phnC 0.71, phnM 0.71). These phn associations are driven by STRING's **genomic-neighborhood** channel and reflect the coincidental physical adjacency of yjdM to the phn operon in *E. coli* — precisely the confounder shown not to generalize across genomes (PMID:40703034). They are circular, not independent functional evidence. The UniProt record P0AFJ1 carries only the keyword "Hydrolase" and, notably, no zinc-binding site feature despite the clear Cys4 ribbon — confirming that the record's current functional identity is entirely downstream of the single disputed study.

Mechanistic Model / Interpretation

The chain of reasoning that produced the GO:0047400 annotation, and where it breaks, can be summarized as follows:



The evidence is best explained by a classic annotation-propagation failure: a **lexical name collision** between two phylogenetically and biochemically unrelated genes, amplified by (a) a machine-learning EC prediction, (b) a single in-vitro assay that a later study could not support in vivo, and (c) a genomic-neighborhood coincidence specific to *E. coli*. None of these constitute direct evidence that the YjdM gene product catalyzes C–P bond cleavage of phosphonoacetate.

The positive signal in the data points elsewhere. The conserved dual-zinc-ribbon architecture and the tetrahedral Cys4 geometry are the strongest, most reproducible structural features of YjdM. These support a **zinc ion binding** molecular function and suggest a nucleic-acid-associated or metal-structural role typical of zinc-ribbon proteins, though the specific biological process remains undetermined.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
PMID:37963869	Computational (ML) + in-vitro assay	Source of hypothesis (weak)	YjdM has phosphonoacetate hydrolase activity	DeepECtransformer predicted EC 3.11.1.2; "experimentally validated … for three proteins (YgfF, YciO, and YjdM)"	*E. coli*, in vitro
PMID:40703034	Review/ reassessment of primary predictions	**Refutes**	Whether YjdM PhnA activity is real	"the PhnA activity observed in vitro is not supported"; lowest confidence (CS=1); yjdM "generally not close to phosphonate catabolism or transport genes" outside *E. coli*	Cross-genome analysis
PMID:9300819	Structural/ evolutionary (sequence)	**Refutes homology**	Is YjdM homologous to true PhnA?	True phosphonoacetate hydrolase "showed no significant homology with any data bank accessions"; ~400-aa novel C–P lyase	*Pseudomonas fluorescens* 23F
PMID:8388873	Mutant/genetic (TnphoA')	**Refutes**	Is yjdM part of the phosphonate operon?	Functional operon is "14 genes named … phnC to phnP" — excludes phnA(yjdM)/phnB	*E. coli*
PMID:2155230	Sequence/ historical		Origin of the "phnA" name	"Seventeen open reading frames (phnA to phnQ) …	*E. coli* B

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
		Refutes (name collision)		in one transcriptional direction" — alphabetical/ positional naming	
PMID:11344133	Structural/ functional (gene region)	Qualifies (true-enzyme context)	Genomic context of the real phnA	Real phnA clusters with phnR (LysR regulator) and phnB (transporter), not zinc-ribbon genes	*Pseudomonas fluorescens* 23F
AlphaFold AF-P0AFJ1-F1	Structural (computational)	**Refutes active site; supports Zn binding**	Hydrolase active site or zinc site?	Cys C6/C9/C23/C26 form tetrahedral Cys4 cluster (SG–SG 3.4–3.8 Å); C93 is 46–49 Å away; mean pLDDT 78.2	Model, 111 aa
STRING 511145.b4108	Interaction/ association (database)	Competing artifact	Does yjdM functionally associate with phn genes?	Top partner yjdN (0.917); phn links (0.71–0.83) via neighborhood channel	*E. coli*

GO Curation Implications

Lead requiring curator verification: Remove, or downgrade and qualify, the **GO:0047400 (phosphonoacetate hydrolase activity)** IDA annotation on P0AFJ1.

- **Molecular Function (the disputed term):** The IDA evidence code is not justified. The sole supporting study (PMID:37963869) is machine-learning-driven with in-vitro-only validation, and it is directly contradicted in vivo by a peer-reviewed reassessment (PMID:40703034).

Recommended actions, in order of preference: (1) **remove** GO:0047400; or (2) if retained provisionally, **change the evidence code away from IDA** to a weaker/qualified code and flag it as in-vitro-only and non-core, with a curator note citing

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- **Better-supported MF lead:** Add **zinc ion binding (GO:0008270)** based on the conserved Cys4 ribbon (AlphaFold tetrahedral geometry; Pfam PF03831/PF08274; InterPro IPR004624). Label as structural/computational inference (ISS/structural) pending experimental confirmation, not IDA.
- **Cellular Component:** Retain **cytosol (GO:0005829)** if independently supported; not in conflict with the above.
- **Biological Process:** No specific BP is supported. Do **not** annotate phosphonate catabolic process. The gene is genetically excluded from the phnC–phnP operon (PMID:8388873).

The net recommendation is that GO:0047400 should **not** be treated as yjdM's function, and yjdM should be curated as a conserved zinc-ribbon protein of **unknown primary function** with a zinc-binding structural feature.

Mechanistic Scope

The immediate molecular function being tested is the direct catalysis of C–P bond cleavage of phosphonoacetate (EC 3.11.1.2) by the YjdM polypeptide. The evidence relevant to that **direct gene-product activity** is: (a) one ML prediction + in-vitro assay that was not reproduced in vivo; (b) the absence of any sequence or structural homology to the true enzyme family; and (c) a structural model showing a zinc-binding ribbon rather than a hydrolase active site. None of these support direct catalytic activity.

The signals that might superficially *suggest* a phosphonate role are all **downstream or contextual**, not direct: the "phnA" name (a lexical artifact), the *E. coli*-specific genomic neighborhood (a gene-order coincidence), and STRING associations (a neighborhood-channel inference derived from that same coincidence). No loss-of-function phenotype, no in vivo growth-on-phosphonate dependency, and no biochemical reconstitution establishes a phosphonate-catabolic role for yjdM.

Conflicts and Alternatives

1. **Name collision (primary alternative explanation).** Two unrelated genes are both called "phnA": the *Pseudomonas fluorescens* 23F phosphonoacetate hydrolase (PMID:9300819; PMID:11344133) and the *E. coli* b4108/yjdM ORF (PMID:2155230). The GO transfer appears to have flowed across this lexical bridge rather than across genuine homology.
2. **Organism-specific genomic context.** The yjdM–phn adjacency is specific to *E. coli*; across the broader IPR004624 family it does not hold (PMID:40703034). Any functional inference from *E. coli* gene order is therefore not generalizable.
3. **In-vitro-only artifact.** The reassessment (PMID:40703034) frames the in-vitro activity as unsupported in vivo — consistent with a spurious or promiscuous low-level activity under non-physiological conditions, a well-known risk when validating ML enzyme predictions.
4. **Database carry-over.** The UniProt "Hydrolase" keyword and the GO IDA together now form a self-reinforcing record even though they derive from a single disputed study; the missing zinc-binding feature annotation shows the record under-represents the best structural evidence.

Alternative interpretation favored by the data: YjdM is a zinc-ribbon protein whose conserved function involves zinc binding and, plausibly, nucleic-acid association (a common role for zinc ribbons), but whose specific biological process is uncharacterized.

Evidence Base

PMID	Title (abbrev.)	Role in this assessment
37963869	DeepECtransformer ML enzyme prediction + validation	Sole source of the GO:0047400 IDA (ML + in-vitro)
40703034	Reassessment of the predictions	Refutes: in-vitro PhnA activity "not supported" in vivo; CS=1
9300819	Cloning of *Pseudomonas* phosphonoacetate hydrolase	True enzyme is non-homologous to YjdM
11344133	phnA gene region analysis in *P. fluorescens* 23F	True enzyme's pathway/regulatory context (phnR, phnB)
8388873	TnphoA' mutational analysis of the phn operon	Functional operon is phnC–phnP; excludes yjdM
2155230	Sequence of the *E. coli* phosphonate region	"phnA" is a positional/alphabetical label (name collision)
21261917	Phosphonoacetate biosensor via PhnR	Orientation on the true phosphonoacetate degradative system

The literature cleanly separates into two groups. The *Pseudomonas* phosphonoacetate-hydrolase papers (PMID:9300819, PMID:11344133, PMID:21261917) describe the authentic EC 3.11.1.2 enzyme — large, non-homologous to YjdM, and embedded in its own phnR/phnB regulatory and transport context. The *E. coli* genetics papers (PMID:2155230, PMID:8388873) show that the *E. coli* "phnA" label is positional and that yjdM sits outside the functional phnC–phnP operon. The two modern papers (PMID:37963869 and PMID:40703034) are, respectively, the origin and the refutation of the annotation under review.

Limitations and Knowledge Gaps

- **True function unknown.** Refuting phosphonoacetate hydrolase activity does not identify what yjdM actually does. The zinc-ribbon architecture suggests zinc binding and possibly nucleic-acid interaction, but no primary literature establishing an alternative function was found. *Resolved by:* targeted biochemistry and genetics (below).

- **AlphaFold is a model.** The Cys4 geometry is derived from a predicted structure (mean pLDDT 78.2), not an experimental structure. The geometry is unambiguous and internally consistent, but experimental confirmation (zinc content, EXAFS, or crystallography) is lacking. *Resolved by:* metal-content analysis and structure determination.
- **In-vitro assay details not independently re-examined here.** We rely on the reassessment's conclusion (PMID:40703034) rather than re-analyzing the raw DeepECtransformer assay data. *Resolved by:* curator inspection of

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supplementary data and

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scoring rationale.

- **Snippet verification status.** Several key quotes (P 40703034, P 37963869, P 2155230) are flagged "unchecked" in the knowledge state; the

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and

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snippets are verified. A curator should confirm the unchecked snippets against the source abstracts.

Discriminating Tests

1. **In vivo phosphonoacetate growth assay.** Test whether a yjdM deletion abolishes, and complementation restores, growth on phosphonoacetate as sole phosphorus source in *E. coli*. The phnC–phnP genetics predict **no** yjdM dependence. Most decisive single experiment.
2. **Purified-enzyme kinetics with proper controls.** Re-measure phosphonoacetate hydrolase activity of purified YjdM against a true PhnA positive control and a catalytically-dead

negative control, with metal-dependence profiling. Distinguishes genuine catalysis from a non-specific in-vitro signal.

3. **Metal-content determination.** ICP-MS / atomic absorption on purified YjdM to confirm stoichiometric zinc, plus mutagenesis of C6/C9/C23/C26 to test zinc dependence of folding/stability. Confirms the Cys4 zinc site.
4. **Cross-genome synteny and phylogenetics.** Formal analysis of IPR004624 members' gene neighborhoods to confirm the *E. coli* phn adjacency is not conserved — operationalizes the

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   observation quantitatively.
5. **Structural biology.** Crystallography or cryo-EM of YjdM (or ortholog) to test the zinc-ribbon model and rule out any hydrolase active-site pocket.

Proposed Follow-up Actions / Curation Leads (require curator verification)

Candidate action change: - GO:0047400 (phosphonoacetate hydrolase activity): **remove** or **downgrade/qualify** — change evidence away from IDA; mark in-vitro-only, non-core; add curator note citing the refuting reassessment.

Candidate new/replacement GO terms: - Add **GO:0008270 zinc ion binding** (MF) as a structural/computational lead (Cys4 ribbon; IPR004624; AlphaFold tetrahedral geometry) — label as ISS/structural, not IDA. - Retain **GO:0005829 cytosol** (CC) if independently supported. - Do **not** add any phosphonate catabolic process (BP).

Candidate references and exact snippets to verify: - PMID:40703034: *"In conclusion, the PhnA activity observed in vitro is not supported, and additional in vivo experiments are required to confirm the biological role of this enzyme."* and *"However, except for E. coli, yjdM genes are generally not close to phosphonate catabolism or transport genes."* - PMID:37963869: *"We experimentally validated the enzymatic activities predicted for three proteins (YgfF, YciO, and YjdM)."* - PMID:8388873: *"All genes for phosphonate (Pn) utilization in Escherichia coli are in a large cluster of 14 genes named, in alphabetical order, phnC to phnP."* - PMID:2155230: *"Seventeen open reading frames (phnA to phnQ) were identified in one transcriptional direction."* - PMID:9300819: *"The nucleotide and deduced amino acid sequences of the phnA gene showed no significant homology with any data bank accessions."*

Suggested curator questions: - Does the GO IDA annotation cite any evidence independent of ?

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(Finding: no.) - Has the in-vitro activity ever been reproduced in vivo? (Finding: no — explicitly unsupported per .)

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- Should UniProt P0AFJ1 gain a zinc-binding site feature (C6/C9/C23/C26)? (Recommended: yes, pending confirmation.)

Suggested experiments: the five discriminating tests above, prioritizing the in vivo phosphonoacetate growth assay in a yjdM deletion background.

Prepared for AI Gene Review hypothesis-level curation. All quantitative and structural claims above are drawn from the cited primary literature, the AlphaFold AF-P0AFJ1-F1 model, and public STRING/UniProt records. Computational results (AlphaFold geometry, STRING scores) are reported as model/database-derived inferences, distinct from direct experimental results.
